Catching Hackers by Watching Watts: A Measurement Study of Power-Only Attack Detection on Consumer IoT Devices

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Abstract. Security analysis of consumer Internet of Things (IoT) devices is increasingly constrained by limited visibility into encrypted traffic and closed firmware, which motivates the exploration of observable and privacy preserving external signals. Prior work has shown that adversarial activity can induce measurable changes in device power usage, indicating that power based intrusion detection is feasible, yet its stability and reliability under realistic operating conditions remain unclear. We address this gap through a systematic measurement study of 33 commercial IoT devices, collecting per second power traces and complete background traffic logs during idle operation and realistic active use under DoS and Reconnaissance attacks. Our analysis shows that power usage provides clear separation between benign and adversarial behaviour in stable idle states, while legitimate activity introduces masking that varies across devices. We identify 4 behavioural regimes that describe how activity suppresses or alters attack related increments and analyse power stability and background traffic characteristics to explain the observed differences in detection strength. Our findings demonstrate that power based detection is viable in real environments and highlight the physical and network conditions that shape the detection reliability.

Keywords: IoT security · Machine learning · Internet measurement.

1 Introduction

Consumer Internet of Things (IoT) devices are becoming more prevalent in households, providing consumers with convenience, connectivity, and automation [23]. Along with these benefits come concerns regarding privacy and security [26,4,12,19]. These devices rely on closed firmware and vendor controlled ecosystems, and they communicate through encrypted channels that restrict the visibility available to local observers [20]. Their behaviour is further shaped by cloud mediated services that operate outside the control of household networks, which complicates the task of understanding device actions and identifying adversarial activity [1,30]. As a result, traditional network based defences that rely on payload inspection or protocol specific analysis provide limited insight, motivating the exploration of alternative observable signals that remain accessible despite encryption and opaque device logic.

One such signal is power consumption. Prior work has shown that adversarial activity can induce detectable changes in device power usage, suggesting that power traces can serve as a side channel for intrusion detection [24]. The initial findings underscore the promise of power-based intrusion

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detection, yet a more reliable and comprehensive evaluation is required to understand how power-based signals behave across this diversity and realistic usage conditions.

Power-based attack detection is feasible, yet its behaviour under realistic operating conditions is not well understood. Consumer IoT devices vary in hardware design, firmware behaviour and cloud dependence, and their power usage is influenced by many factors that extend beyond adversarial activity. User-driven actions introduce additional workload, and background traffic generated by vendor services creates continuous variation in device behaviour. These sources of variability can alter or mask the increments associated with attacks. Moreover, their influence differs across devices and across usage states. As a result, the stability, reliability and generalisability of power-based detection across real devices, real activity patterns, and real network conditions remain open questions. This situation highlights the need for systematic measurement to determine when power-based detection performs reliably and when it is challenged by the surrounding workload.

In this work, we investigate how network attacks manifest in the power consumption of consumer IoT devices under both idle operation and realistic active use. We evaluate 33 commercial devices across five categories and collect per second power traces together with full background traffic while launching four categories of attacks. Using a machine learning classifier based on Random Forests [6], we show that power usage alone contains sufficient statistical structure to reliably distinguish attack and benign scenarios across a wide range of devices. We further demonstrate that legitimate user activity introduces non-uniform masking effects that substantially degrade detection for some devices while leaving others largely unaffected. By jointly analysing power signals and background traffic, we identify the conditions under which power based detection remains effective and explain the factors that limit its reliability in realistic deployments. Our key contributions are as follows:

- We conduct the first large scale empirical study of power based attack detection on 33 commercial IoT devices and show that common network attacks can be detected from power traces alone with consistently high accuracy across multiple device categories.
- We demonstrate that power side channel detection performance varies significantly between idle and active states, and quantify how legitimate user activity degrades detection accuracy in a device and attack dependent manner.
- We identify 4 distinct masking regimes induced by user activity and show that these regimes explain the heterogeneous robustness of power based detection across devices.
- We trace detection failures to increased power signal instability and show that encrypted communication and service discovery traffic are the primary network level drivers of this effect.

To support further research, all software and data we produced as part of this work are publicly available at `https://github.com/SafeNetIoT/PowerIDS`

2 Threat Model and Research Questions

2.1 Threat Model

We consider a typical smart home security setting in which IoT devices are connected to the local WiFi network in order to access external services on the Internet. The following model outlines the victim, the adversary, and the attack behaviours in our study.

Victim. The victim is a consumer IoT device that operates under normal user activity, interacts with cloud services and peer devices, and exposes typical vulnerabilities present in commodity IoT products. The victim may also be an individual who owns, uses, or benefits from such consumer IoT devices.

Adversary. The adversary is any party that can access the IoT device traffic and launch attacks. Examples include external security threats, on-path network observers, third parties, other IoT devices in the environment.

Threat. We consider security threats that lead to increased network activity and power consumption in the device. This includes Denial of Service (DoS) attacks such as ICMP flood and SYN flood, which overwhelm IoT devices by injecting excessive volumes of traffic and can cause significant power spikes or device malfunction [17,32,22]. We also consider reconnaissance activity such as Port scanning and OS fingerprinting, which systematically probe device interfaces and expose system characteristics, enabling targeted follow-up attacks while inducing measurable shifts in network processing and power consumption.

2.2 Research Questions

The main goal of this work is to determine whether network attacks can be identified from power traces alone and to understand the conditions that shape this capability across diverse IoT devices. In particular, this work answers the following research questions.

RQ1. Detection Feasibility. *Can distinct network attacks such as DoS and reconnaissance be detected and classified solely through power consumption patterns on resource-constrained IoT devices?*

This question targets the fundamental capability of the system. It evaluates whether power traces alone contain enough discriminatory structure to distinguish normal behaviour from attack behaviour. The attack implementations include ICMP flood, SYN flood, port scanning and OS fingerprinting. The goal is to establish the upper bound of what power based detection can achieve under ideal conditions, before considering workload interference or device-side complexity.

RQ2. Robustness to Activity. *How significantly does legitimate user activity degrade detection performance compared to when the device is idle?*

This question examines the *Masking Effect* created by device activity. Idle devices usually maintain stable and low-variance power usage, but active states such as video streaming or audio playback introduce large fluctuations and irregular swings that form a source of power instability. These fluctuations can conceal the small increments produced by network attacks. Our analysis reveals four distinct masking behaviours which determine whether activity moderately degrades detection performance, improves it, or causes collapse.

RQ3. Performance Verification. *To what extent do intrinsic device behaviors, specifically power-signal stability and background traffic, correlate with detection failure?*

This question explains why power-based detection succeeds for some devices and fails for others. The analysis first characterizes power stability through the Mean Absolute Successive Change metric, which captures how much a device's power draw fluctuates during normal operation and therefore how cleanly attack-induced increments can appear above the noise floor. It then examines background traffic generated by the device, including cloud heartbeat and keep alive flows, media related TLS or UDP streams, system telemetry bursts and local discovery protocols such as mDNS

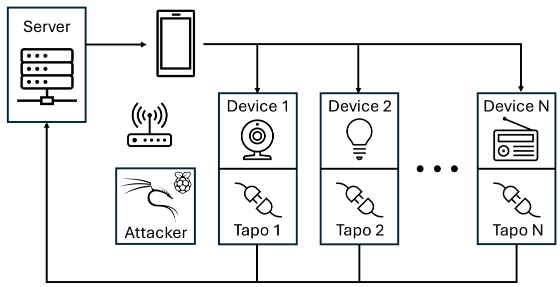

Fig. 1: Overview of the IoT testbed.

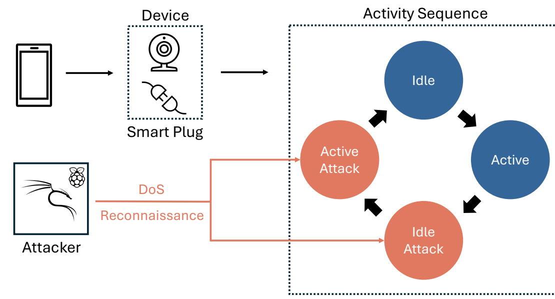

Fig. 2: Overview of the data collection methodology

and SSDP. These network behaviours shape both packet visibility and internal workload dynamics, which in turn influence the power trace. The goal is to determine how the interaction between power instability and background traffic intensity aligns with observed detection failure.

3 Testbed

Our experiments take place in a controlled environment designed to capture the joint evolution of power traces and background network traffic across a diverse set of consumer IoT devices. As shown in Figure 1, the testbed consists of a central gateway, consumer IoT devices connected via WiFi, per-device smart plugs for power monitoring, four Raspberry Pi nodes acting as internal attackers, and an Android phone used to drive device interactions. This setup enables repeatable execution of device activities and attack scenarios under consistent and realistic operating conditions.

Devices and Activities. We consider consumer IoT devices typically deployed in a smart home. Our evaluation includes 33 consumer IoT devices spanning five categories: lights, audio assistants, cameras, home appliances, and televisions, as summarised in Table 1. The idle state reflects baseline connectivity with no user interaction, while the active state corresponds to the primary user driven function for that device category, such as audio playback, video streaming, or appliance operation. The selected devices originate from multiple vendor ecosystems and cloud backends, providing diverse firmware behaviours, protocol usage, and power profiles representative of real world smart home deployments.

Attackers. Adversarial activity is generated by four Raspberry Pi nodes running Kali Linux and located on the same network as the IoT devices. We consider two classes of attacks. Denial of service (DoS) attacks include SYN flooding and ICMP flooding, while reconnaissance attacks include port scanning and operating system fingerprinting. We execute attacks in controlled cycles that interleave attack and recovery phases, allowing us to repeatedly observe the impact of adversarial traffic on both network behaviour and power consumption across idle and active device states.

Measurement and Synchronisation. Device power consumption is monitored using per-device TP-Link Tapo P110 smart plugs [34] through the Tapo API [9] that report active power measurements

Table 1: Consumer IoT devices and associated user activity scenarios

Category	**Lights**	**Audio**	**Camera**	**Appliance**	**TV**
Devices N = 33	Yeelight Lifx Mini Wiz Smart Bulb Lepro Smart Bulb Antela Smart Bulb Tapo L530e Fitop	Sonos Google Home Bose Echo Dot 3 Echo Dot 5 Echo Spot Nest Mini 2 Homepod	Google Nest Cam Amcrest Blurams Security Cam Yi Cam Boifun Baby Cam Hualai Cam	Levoit Air Purifier Honeywell Thermostat Swan Alexa Smart Kettle Xiaomi Blender Weekett Kettle Cosori Air Fryer	LG 32" TV Apple TV A1625 TCL Roku TV Samsung 32" TV Samsung Chromecast Smart TV Toshiba TV
Purpose	Lighting devices controlled through mobile applications	Smart speakers offering voice assistant capability	Devices providing remote monitoring and video capture	Home appliances with remote or automated control	Streaming devices controlled through HDMI
User Activity	Idle: Switch off Active: Switch on	Idle: No audio output Active: Play audio	Idle: No live view Active: Streaming	Idle: Turn idle Active: Function running	Idle: Screen saver or Standby Active: Playback

at one hertz. Network traffic is captured at the gateway, which provides per-device packet-level visibility. An Android phone connected to the gateway is used to trigger device actions through standard companion applications. Power logging and traffic capture operate concurrently throughout each experiment, producing aligned traces that associate network activity with the corresponding power behaviour across idle, active, and attack scenarios.

4 System Design

We introduce a reproducible measurement-driven system [3] designed to determine whether attack activity can be recognised from power traces alone and to understand the network conditions that shape detection outcomes. The system is organised into three stages. First, we establish a controlled data collection testbed that records power traces together with ground-truth network traffic across diverse operational scenarios. Second, we process these traces using a lightweight machine learning pipeline that applies statistical feature extraction and sliding window segmentation to identify attack-related patterns in the power signal. Finally, we examine the surrounding network environment, with a focus on background traffic and activity induced masking, to understand the external factors that influence detection performance and to explain the observed results.

4.1 Data Collection

We collect power and network data through a repeatable measurement procedure illustrated in Figure 2. Each device is evaluated under controlled activity and attack conditions by combining continuous power measurements with gateway level packet captures similar to previous studies [26].

Scenario Design. For each device, we consider four scenarios that combine user activity and adversarial traffic. In the *idle* scenario, the device remains powered and connected to the Internet without user interaction. In the *active* scenario, the primary user function for that category is triggered as listed in Table 1, while no malicious traffic is present. In the *idle attack* scenario, the device stays idle while an attacker targets it with SYN flood, ICMP flood, Port scanning and OS fingerprinting traffic. In the *active attack* scenario, the same set of attacks is launched while the device performs its active behaviour.

[3] `https://github.com/SafeNetIoT/PowerIDS`

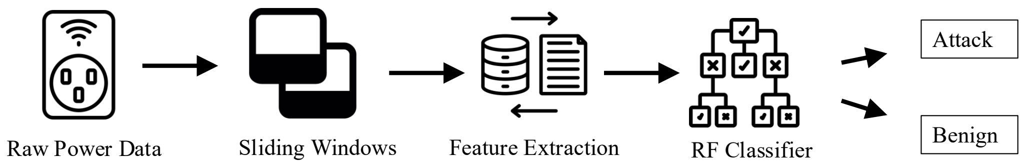

Fig. 3: Machine Learning Pipeline overview: (1) Raw power data captured at 1HZ is converted to sliding windows. (2) Each window goes through statistical feature extraction. (3) RF-based ML Classifier is trained and tested.

Measurement Setup. Each device is exercised for 10 rounds in every scenario. In attack rounds, the attacker generates traffic for 1 minute, followed by a 3 minutes recovery interval before the next round. Benign idle and active rounds follow the same timing so that all conditions share a comparable duration. Throughout all rounds, the Tapo P110 smart plugs record continuous power traces at 1HZ, while complete network packet traces are captured at the gateway from the WiFi interface of the server using full packet capture. Raspberry Pi boards running Kali Linux act as the attackers and run `hping` for SYN and ICMP floods and `nmap` for port scanning and OS fingerprinting experiments.

4.2 Machine Learning Pipeline

To explore RQ1 using power measurements alone, the system applies a supervised classification pipeline that converts raw per-second power readings into short overlapping windows, summarizes each window with simple time-domain statistics, and feeds the resulting feature vectors into a lightweight per-device classifier. An overview of the pipeline is shown in Figure 3. This section describes the sliding-window representation, the feature extraction step, and the classifier and training procedure used throughout the evaluation.

Sliding-Window Segmentation. We transform each device's continuous power signal, sampled at 1 Hz, into a sequence of overlapping windows to capture short-term behavior. Each window spans $W = 30$ seconds of power readings, and the start times of successive windows are separated by $S = 10$ seconds, so adjacent windows share 20 out of 30 samples. This representation preserves local temporal structure while providing multiple views of the same physical period, which is helpful when attacks start or end inside a window.

For every window, we assign a binary label indicating whether it predominantly contains attack activity. Let t_{attack} denote the number of seconds inside the window that occur during an attack. We label the window as “attack” if at least half of its samples are under attack and “benign” otherwise.

Feature Extraction. Given a windowed segment of the power time series, the system computes a compact feature vector that summarizes the temporal structure of the signal.

Table 2: Ablation results comparing alternative feature sets.

Feature Set	**Mean Accuracy (%)**	**Mean F1 (%)**
Baseline	84.87 ± 8.32	76.75 ± 12.33
Temporal Set	83.89 ± 7.445	75.87 ± 10.55
Spectral Set	83.31 ± 9.02	75.19 ± 11.41

Time-Domain Statistics. From each 30 s window, we derive a compact five-dimensional feature vector that summarizes the distribution of power values. Concretely we compute the mean, standard deviation, minimum, maximum, and median. These statistics capture the central level of consumption, its variability, and the presence of short spikes. In Section 5, we show that this minimal feature set already matches the performance of richer alternatives while remaining substantially lighter-weight.

Feature Set Ablation and Justification. To determine the most appropriate feature representation for our system, we conducted an ablation study comparing our compact 5-dimensional statistical feature set against richer temporal and spectral alternatives. Table 2 summarizes the results across the full set of devices.

Across all devices, the baseline 5-feature configuration achieves the best performance, yielding 84.29% mean accuracy and 76.71% mean F1 across all devices. Augmenting the representation with temporal descriptors (MASC, autocorrelation, zero-crossing rate, and related measures) slightly *reduces* overall performance to 83.89% accuracy. Spectral features (centroid, bandwidth, flatness, entropy, dominant frequency) further decrease accuracy to 83.31%, despite their higher computational cost. These spectral features are calculated by first performing a Fast Fourier Transform and then calculating the Power Spectral Density.

These results suggest that richer feature sets do not improve detection accuracy and can even degrade generalisation. Power traces in our setting are short (30-second windows) and have low sampling rates (1 Hz), which limits the discriminative value of higher-order temporal or spectral structure.We therefore adopt the 5-feature statistical representation as the final design choice richer alternatives.

Random Forest Classifier. We treat attack detection as a supervised binary classification problem at the window level and train a separate model for each device. After benchmarking several classical models (logistic regression, support vector machines and small neural networks), we adopt a Random Forest (RF) classifier as the default.

Random Forests handle nonlinear decision boundaries and heterogeneous feature distributions without requiring feature normalisation or complex hyperparameter tuning.

Unless otherwise stated, each device uses an RF with 50 trees, Gini impurity as the split criterion, unlimited depth (trees expand until purity), and class_weight="balanced" to compensate for the smaller number of attack windows. For each device we split its window set into 80% training and 20% validation data, stratified by label, and repeat this process 10 times with different shuffles. We split the train and validation sets sequentially to ensure that there is no data leakage and that the model does not see almost identical windows as we may observe on a random split. We report the mean accuracy and F1 score, focusing on F1 due to class imbalance.

4.3 Activity-Induced Masking Mechanism

During active operation, IoT devices execute additional workloads such as cloud sync, encrypted streams, media playback or local discovery. These behaviours introduce structured fluctuations in both power draw and background traffic. *Televisions* stream online video content such as YouTube through their native applications, which produces sustained encrypted media flows together with periodic bursts driven by adaptive streaming. *Cameras* activate live view mode and send continuous video streams to the companion application on the Android phone, generating stable uplink traffic and increased local processing demand. *Audio* devices play music or broadcast spoken content, which triggers cloud mediated control traffic and steady audio playback pipelines. *Home appliances* operate their primary functions, including constant temperature mode for smart kettles, heating cycles for air fryers and regulation cycles for thermostats, all of which create distinct electrical and network level signatures. *Smart lights* simply switch to their illuminated state and remain relatively stable, producing only modest power increments and occasional control traffic.

These examples illustrate typical activity patterns for each category, but the behaviour of individual devices does not always align cleanly with their category. Devices within the same category can differ substantially in cloud service design, local processing, media handling or update frequency, which means that category level expectations do not fully explain the observed masking behaviour. This variation motivates the background traffic analysis in RQ2 and leads to a behavioural grouping where devices naturally fall into four regimes according to how activity modulates detectability. The system design therefore considers the interaction between activity and attack induced energy rather than treating activity as random noise.

4.4 Background Traffic Analysis

We treat background traffic as the network activity that surrounds a device during normal operation and is independent of any injected attack. This activity reflects routine communication with peer devices, service discovery, cloud synchronisation and other application behaviours. To capture these conditions, we extract packet traces from pure idle and active phases and process them separately in order to obtain a clean view of the network environment in each state. Each pcap is converted into a structured packet record using `tshark`, which exports timestamps, IP addresses, packet sizes, protocol labels and transport ports. The device IP is then identified from the dominant unicast address in the trace, enabling separation of inbound and outbound traffic and consistent computation of flow level statistics.

The resulting features in Table 3 summarise key aspects of the traffic stream, including volume patterns derived from per-second packet rates, packet size behaviour, timing structure from inter arrival gaps, protocol composition inferred from ports and protocol tags, flow organisation and directional balance. These indicators form a compact description of the external conditions under which each measurement is taken, and they are later used for further evaluation.

5 Evaluation

We evaluate the system across the three research questions of this study. RQ1 tests the feasibility of detecting network attacks purely from power consumption signals. RQ2 quantifies the robustness of this approach under legitimate user activity and identifies the conditions under which masking arises. RQ3 evaluates the physical stability of the power trace, quantifies how background traffic

Table 3: Background traffic metrics extracted from idle and active pcaps.

Metric Group	**What is Measured from PCAP**
Volume and variability	Packets per second (PPS), PPS variance, packet count, burst count
Packet size behaviour	Mean packet size, variance, kurtosis
Timing structure	Inter-arrival time mean and variance
Protocol composition	Proportions of TLS, TCP, UDP, mDNS, SSDP, ARP, DHCP
Flow structure	Number of flows, packets per flow, top flow fraction, flow entropy, number of destination IPs and destination ports
Direction and strength	Inbound to outbound ratio, background traffic strength (BTS)

Table 4: Attack detection performance across device categories .

Category	**Accuracy (%)**	**F1 (%)**
Appliances	89.6 ± 4.0	85.2 ± 6.1
Speakers	80.7 ± 4.3	72.4 ± 7.6
Cameras	85.5 ± 8.2	79.1 ± 12.1
Smart Bulbs	90.8 ± 4.9	87.8 ± 6.3
TVs	79.2 ± 7.5	67.5 ± 14.3
Overall Mean	84.9 ± 8.3	76.8 ± 12.3

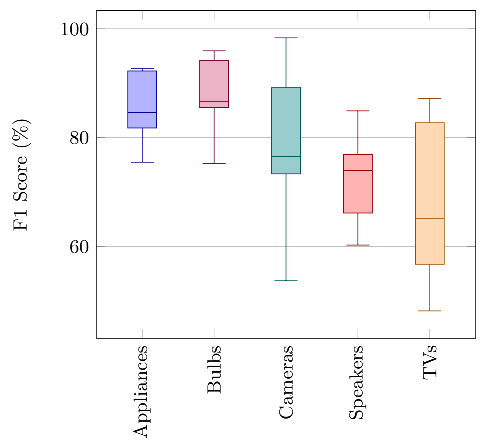

Fig. 4: Distribution of F1 scores across device categories.

behaviours influence detection, and examines how different attack characteristics interact with these behaviours to shape performance.

5.1 Detection Feasibility (RQ1)

RQ1 asks whether attack activity can be detected from power consumption alone. We evaluate this by training the Random Forest classifier using *idle*, *idle-under-attack*, *active* and *active-under-attack* windows. This setup allows us to answer if observing the fluctuations in power-draw caused by attacks on an IoT device are enough to accurately detect these scenarios.

For each device, we compute the accuracy and F1 score on a held-out validation set. All results in this section use the compact statistical feature set and the per-device RF configuration described in Section 4.2. Individual device results in this section can be found in Appendix A.

Overall Performance. Table 4 summarizes the results across all the devices. Across the 33 devices, our system achieves a mean accuracy of 84.9%, with categories like smart bulbs and appliances

Table 5: Average drop in attack detection accuracy and F1 from idle to active.

Category	*Δ*Accuracy (%)	*Δ*F1 (%)
Appliances	6.82	11.97
Smart Bulbs	4.95	7.61
Cameras	4.88	5.45
Speakers	14.54	21.55
TVs	15.38	23.43

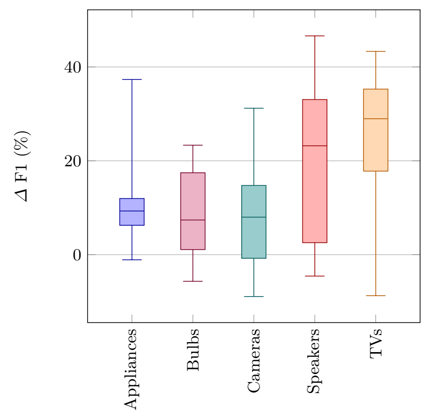

Fig. 5: Distribution of F1 drop (*Δ*F1) when shifting from idle to active windows.

resulting in 90% accuracy. These results demonstrate that power traces contain sufficient structure to reliably separate attack from benign behavior in stable operating conditions.

Figure 4 represents the F1 scores of the devices within the categories. Generally, we see that categories that we assume to have less variability (e.g., bulbs) within their default power-draw signal have higher F1 scores compared to devices that have higher variability due to the device function (e.g., TVs). This implies that the nature of the device and its activity play a role in masking its power signal from fluctuations that help us detect attacks. Nevertheless, power draw values are a good predictor for detecting attack scenarios on IoT devices.

Idle-Active Difference. To isolate the effect of activity on its power draw signal, we analyze the difference in attack accuracy on these devices between their idle and active scenarios. This approach allows us to see if the activity on the device causes instability within the power draw of the device that can mask the fluctuations caused by an attack.

Table 5 shows that appliances and smart bulbs experience only modest degradation when moving from idle to active states (4–7%), reflecting their stable power profiles. Cameras also show relatively small drops on average (3.4%), although individual cameras vary widely.

In contrast, speakers and TVs exhibit substantially larger losses (13–16%), indicating that user activity introduces power fluctuations that reduce separability between benign and attack windows. Section 5.2 explores how these drops correlate with each device's background traffic and network intensity.

Analyzing the box-plot in Figure 5 allows us to see that, in general, the idle scenario is much easier to classify than the active scenario for all devices. For TVs and Speakers this is more significant and implies that the lower general performance is due to the impact from their activity scenarios. We also see that there is great variance within these categories, which motivates us to quantify this masking approach with different metrics. This question is further explored in Section 5.3 where we define a Mean Absolute Successive Change metric to determine default stability of a device and correlate this with the performance of the system. Detection performance is greatly impacted by the type of device as this directly correlates with how strongly the power draw signal can be masked.

Table 6: Attack-wise detection accuracy across device categories (mean across devices).

Category	**ICMP %**	**OS %**	**SYN %**	**PORT %**
Appliances	91.9 ± 4.3	88.9 ± 6.3	87.9 ± 5.3	89.9 ± 4.5
Speakers	78.6 ± 7.3	81.3 ± 9.6	80.2 ± 6.2	83.0 ± 7.9
Cameras	86.9 ± 9.0	83.1 ± 9.4	82.8 ± 10.4	89.3 ± 8.3
Smart Bulbs	95.1 ± 2.3	89.1 ± 6.7	86.4 ± 11.2	92.7 ± 4.6
TV	80.0 ± 8.1	77.1 ± 9.1	79.4 ± 6.1	80.1 ± 11.5

Per-Attack Performance. While the previous analysis aggregates all malicious activity into a single attack class, different network attacks generate distinct traffic patterns that may produce different power signatures. To understand how sensitive the system is to individual attack families, we evaluate detection performance separately for ICMP flooding, OS fingerprinting, SYN flooding, and port scanning attacks. Table 6 summarizes the mean accuracy for each attack type across device categories.

Across all categories, ICMP flooding is the easiest attack family to detect, yielding the highest accuracy most categories. While Port scans also achieve strong performance.

In contrast, OS fingerprinting and SYN flooding yield systematically lower detection accuracy, especially for speakers and TVs. These devices exhibit more dynamic and user-dependent power profiles, making their fine-grained power fluctuations harder to distinguish from the small perturbations caused by lightweight reconnaissance behavior. This aligns with the masking effects observed in Section 5.1.

Takeaway. Detection performance is affected both by the default activity of the device and what attack is being performed. Per-device, some attack classes are significantly easier to classify than others; however, these classes don't generalize over the entire testbed.

5.2 Robustness to Activity (RQ2)

Following the observed idle to active performance differences, we investigate how legitimate activity masks attack induced power variations. In RQ2, we quantify this masking effect by grouping devices based on their performance changes under activity and relating these groups to background traffic structure.

Behavioural Clustering. We group devices based on their activity induced performance change. For each device, we compute idle and active F1 scores averaged across all attack families and derive a behavioural signature consisting of idle F1, active F1, absolute drop, relative drop, and rank shift. After standardisation, we project these signatures using principal component analysis (PCA) and cluster devices into four behavioural groups using k-means. Figure 6 shows that these groups form well separated regions in the projected space.

Performance Impact Across Groups. The four behavioural groups exhibit markedly different sensitivity to activity. Figure 7 and Table 7 summarise the corresponding performance changes.

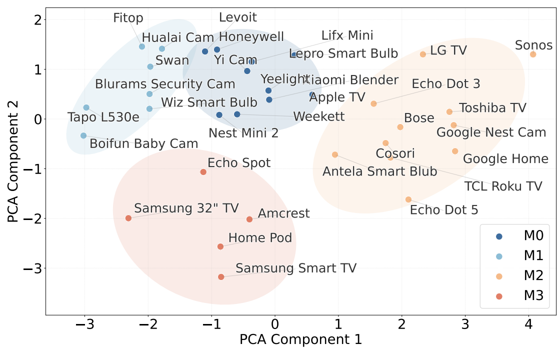

Fig. 6: PCA projection of the behavioural signature space showing the four masking behaviour groups. The ellipses illustrate the 95% confidence region for each group and highlight the clear separation between the clusters.

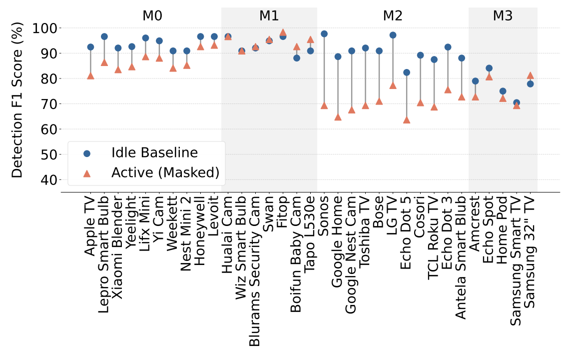

Fig. 7: Idle and active detection performance across masking behaviour groups. Lines connect the idle and active F1 scores for each device and illustrate the strength of the masking effect within each group.

Table 7: Detection performance across masking behaviour groups showing idle and active F1 scores as well as absolute and relative drops.

Cluster	Idle F1 (%)	Active F1 (%)	Abs Drop (%)	Rel Drop (%)
M0	93.96 ± 2.41	86.75 ± 3.91	7.21	7.70
M1	92.86 ± 3.24	94.56 ± 2.60	-1.70	-1.90
M2	90.63 ± 4.33	70.04 ± 4.11	20.59	22.70
M3	77.27 ± 5.04	75.23 ± 5.40	2.04	2.60

Devices in M1 remain largely unaffected by activity, with negligible performance change or slight improvement. M0 shows moderate degradation with the drop between 5%–10%, while M2 experiences substantial masking, with mean F1 dropping by over 20% points under activity. M3 exhibits lower baseline performance and small but consistent decreases with idle around 77% and a mean drop around 2%. These results demonstrate that masking is not uniform across devices and that activity can severely degrade detection for a subset of behaviours.

Feature Structure Verification. To understand the structural differences behind the four behavioural groups, we examine group-level feature patterns. Figure 8 presents a heatmap of the most discriminative features across groups, where values represent Z-score deviations from the global mean. The observed patterns align with the performance trends:

– **M0(quiet devices)** show mostly negative values with low packet rate, low flow count and low variance. Idle and active remain similar, which match the moderate reduction.
– **M1(stable devices)** stay near zero across most features. Idle and active behaviours are almost identical, with only small increases in flow level structure. This aligns with the stable performance and slight improvement under activity and the reinforced pattern in this group.
– **M2(affected devices)** show the strongest difference between idle and active states. Active behaviour presents larger packet sizes in both mean and variance, a higher share of TLS traffic,

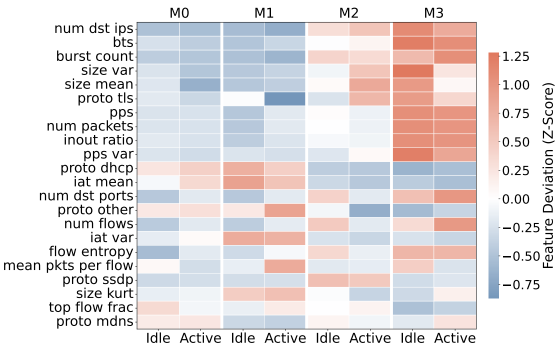

Fig. 8: Group level idle and active feature patterns for the four behaviour groups (M0-M3). Shown are the most discriminative features, Z scored across all devices.

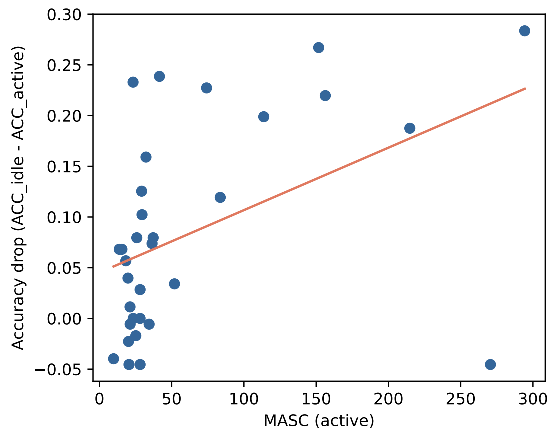

Fig. 9: Relationship between active state power instability ($\mathrm{MASC}_{\mathrm{active}}$) and the drop in detection accuracy from idle to active, after removing the three largest MASC values.

and a reduction in destination ports, flow counts and flow entropy. This reflects a shift toward heavier but more concentrated traffic.

– **M3(noisy devices)** show a consistently high noise floor, with strong volumetric and variability features such as pps, burst count, packet counts and flow activity in both idle and active states. This stable level of intensity limits the separation between states and reflects the weaker baseline behaviour of this group.

Takeaway. Legitimate activity induces heterogeneous masking effects across devices, power-based detection remains reliable for groups whose structure stays stable across idle and active states.

5.3 Performance Verification (RQ3)

To verify the mechanisms underlying activity-induced performance degradation, we examine the relationship between power signal stability, background traffic structure, and detection accuracy.

Power Stability. To quantify how stable each device is under normal operation, we use a time-domain volatility measure, the *Mean Absolute Successive Change* (MASC). For a power trace $\{p_t\}_{t=1}^{T}$ within a window, we define

$$\mathrm{MASC}(p) = \frac{1}{T-1}\sum_{t=2}^{T}\left|p_t - p_{t-1}\right|.$$

Higher MASC values indicate more irregular and fluctuating power traces, while lower values reflect stable behaviour. For each device, we compute the idle and active MASC by averaging over all corresponding windows. We then examine whether the power instability during activity

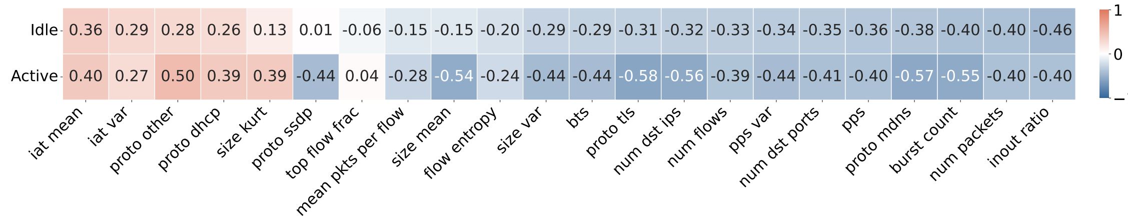

Fig. 10: Correlation of Background Traffic Indicators with Detection Performance in Idle and Active States

predicts the loss in detection performance. Across all devices, active-state MASC shows a strong monotonic association with the drop in detection accuracy from idle to active operation (Spearman $\rho = 0.54$, $p < 10^{-3}$), while the linear association is weak (Pearson $r = 0.18$, $p = 0.31$) due to a small number of extreme outliers. After excluding these outliers, the linear relationship becomes statistically significant (Pearson $r = 0.45$, $p = 0.0095$), while the monotonic association remains high (Spearman $\rho = 0.52$, $p = 0.0025$). Figure 9 illustrates that, for the majority of devices, detection accuracy decreases smoothly as active-state power instability increases.

Global Influence of Background Traffic. To identify the factors underlying detection failure, we compute Spearman correlations between background traffic indicators and detection performance across devices. Figure 10 reveals two consistent feature categories when comparing idle and active states. We identified two categories of features that influence performance: *robust predictors* that remain effective across both states and *masking mechanisms* that become influential during activity.

Indicators such as discovery activity, burstiness, port diversity, timing regularity, and packet intensity act as robust predictors and consistently correlate with detection performance in both idle and active conditions, indicating that these behaviours define a persistent background structure. In contrast, features related to encrypted traffic and service discovery function as masking mechanisms, showing markedly stronger negative associations in the active state and demonstrating that activity amplifies existing traffic structure and increases the effective noise floor rather than introducing random variation.

Attack-Specific Vulnerability Patterns. We correlate each attack-specific F1 score with active background traffic indicators in order to characterise how different attack behaviours influence detection. The results in Figure 11 reveal that each attack exhibits a distinct vulnerability profile shaped by the structure of the surrounding traffic. *ICMP Flooding* are primarily masked by network complexity, with discovery traffic and destination diversity acting as the strongest inhibitors (e.g., proto mdns, $r = -0.70$). In contrast, *Port Scanning* are sensitive to payload dynamics, where legitimate large packet activity (e.g., size mean, $r = -0.57$) and encrypted traffic (proto tls) dilute the statistical footprint of small scanning packets. Protocol-oriented attacks such as *SYN Flooding* and *OS Fingerprinting* exhibit a unique positive correlation with DHCP traffic ($r = +0.45$), indicating improved detectability in simpler, maintenance-heavy environments. Full correlation results are presented in Appendix B.

Takeaway. Detection performance is shaped jointly by power variability, background traffic structure and the behaviour of each attack.

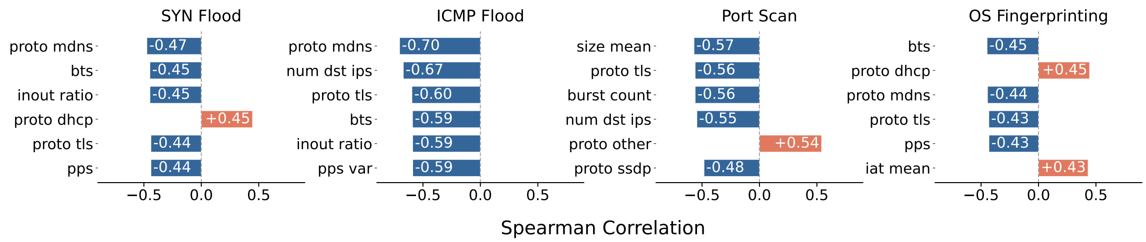

Fig. 11: Top active-state features correlated with detection performance across 4 attack types, where blue bars denote masking features (negative correlation) and orange bars denote helper features (positive correlation).

6 Related Work

There are broadly two areas of work relevant to this paper: *(i)* anomaly/threat detection methods for IoT applications and services, and *(i)* analysis of power usage from consumer IoT devices.

6.1 IoT Anomaly Detection Methods

A large body of work applies time series analysis and machine learning techniques to IoT anomaly and threat detection [33,31,2,13,15]. Most existing systems operate on network traffic features and employ supervised or unsupervised learning to identify malicious behaviour [8,5,20,25,27,28,35]. While effective, these approaches increasingly face limitations due to encrypted communication [11], cloud mediated device behaviour, and restricted traffic visibility in home networks.

Several recent works explore privacy preserving network side analysis. Safronov et al. [30] combine machine learning with rule based filtering on home routers, while HoMonit [37] and Horus-Eye [10] infer device behaviour using packet timing, size patterns, and multi stage analysis pipelines. Although these systems reduce reliance on payload inspection, they still depend on network traffic observability and often require a large set of features, which imposes computational overhead on resource constrained deployments [3,7,36].

6.2 IoT Power Analysis Methods

Power consumption has been explored as a side channel for analysing IoT behaviour and detecting attacks. Prior studies have shown that power traces can reveal DDoS activity, botnet behaviour, and other attacks on embedded platforms such as Raspberry Pi and DragonBoard devices [24,16,18]. A recent survey [21] highlights growing interest in energy based intrusion detection, but also notes that most evaluations rely on small laboratory testbeds with limited device diversity.

Other work has explored the relationship between power usage and communication patterns, for example by analysing power consumption trends in smart city systems [14] or battery drain in edge devices under packet flooding attacks [29]. However, these studies focus on energy impact rather than detection and do not examine the separability between benign activity and attacks.

In contrast to prior research, we conduct a large scale measurement study on a diverse set of IoT devices representative of real home deployments. We systematically evaluate power based attack detection under both idle and realistic active conditions, and analyse how background traffic and device behaviour affect detection reliability. To the best of our knowledge, this is the first study to characterise the feasibility and robustness of power based intrusion detection across a broad range of consumer IoT devices and usage patterns.

7 Discussion

ML and Hybrid Designs. The aggregate ML results in Section 5 show that short power windows summarised with statistical features and per-device Random Forest classifiers are sufficient to detect attacks across a broad range of devices. At the same time, our analysis of idle–active drops, masking groups, and the MASC stability metric highlights that this capability is not uniform: speakers and media devices with highly volatile active-state power traces or complex background traffic require more care, and power-only detection can degrade significantly in these regimes. Together, these findings can suggest a natural hybrid design in which power analysis provides a first line of defence, and network-side features are selectively engaged only for devices or activity regimes that exhibit high instability or adverse background patterns.

Influence of User Activity. As illustrated in RQ2, masking arises because devices do not respond uniformly when user behaviour is introduced. The study defines idle as an unused condition and active as a representative use case for each device, which allows us to measure a consistent baseline of activity induced change. Real households involve a wider range of behaviours that may create different patterns. These variations are not evaluated here, and future work can expand the activity space to capture the full diversity of user interaction.

Temporal Evolution and Firmware Variation. Power and traffic behaviour may evolve over time as firmware and cloud services change, and our dataset therefore represents a snapshot of current device configurations. Such longitudinal variation can shift power and traffic patterns and affect classification boundaries, suggesting that periodic retraining may be needed to maintain accuracy. Importantly, our results indicate that masking is driven by structural device behaviour, implying that the relative robustness patterns observed are likely to persist despite temporal drift.

8 Conclusion

In this paper, we presented a extensive measurement study characterizing the feasibility and robustness of detecting network attacks on IoT devices solely through power consumption side channels. By analyzing a diverse set of 33 consumer IoT devices across 5 categories against SYN Flood, ICMP Flood, Port Scaning, and OS Fingerprinting, we found that while detection is generally effective in idle states, achieving a mean F1 score of roughly 85%, legitimate user activity introduces varying levels of performance degradation. We demonstrate that using a Random Forest based machine learning pipeline on a sliding window approach allows us to accurately determine attack scenarios. While stable devices like smart bulbs remain robust, complex media devices experience significant drops due to the masking effect of their active power profiles. We showed that this degradation correlates directly with physical power instability driven by background behaviors such as encrypted streaming and discovery protocols. Ultimately, our results demonstrate that power based detection is a viable and lightweight approach under realistic conditions, while its reliability is shaped by the interaction between device stability, user activity and surrounding network behaviour.

References

1. Acar, A., Fereidooni, H., Abera, T., Sikder, A.K., Miettinen, M., Aksu, H., Conti, M., Sadeghi, A.R., Uluagac, S.: Peek-a-boo: I see your smart home activities, even encrypted! In: Proceedings of the 13th ACM Conference on Security and Privacy in Wireless and Mobile Networks. pp. 207–218 (2020)
2. Al-amri, R., Murugesan, R.K., Man, M., Abdulateef, A.F., Al-Sharafi, M.A., Alkahtani, A.A.: A review of machine learning and deep learning techniques for anomaly detection in iot data. Applied Sciences **11**(12), 5320 (2021)
3. Alahmadi, A., Aljabri, M., Alhaidari, F., Alharthi, D., Rayani, G., Marghalani, L., Alotaibi, O., Bajandouh, S.: Ddos attack detection in iot-based networks using machine learning models: A survey and research directions. Electronics **12**, 3103 (07 2023). `https://doi.org/10.3390/electronics12143103`
4. Alrawi, O., Lever, C., Antonakakis, M., Monrose, F.: Sok: Security evaluation of home-based iot deployments. In: Proc. IEEE Symp. on Security and Privacy. pp. 1362–1380 (2019)
5. Antonakakis, M., April, T., Bailey, M., Bernhard, M., Bursztein, E., Cochran, J., Durumeric, Z., Halderman, J.A., Invernizzi, L., Kallitsis, M., et al.: Understanding the mirai botnet. In: 26th USENIX security symposium (USENIX Security 17). pp. 1093–1110 (2017)
6. Breiman, L.: Random forests. Machine learning **45**(1), 5–32 (2001)
7. Celik, Z.B., Tan, G., McDaniel, P.D.: Iotguard: Dynamic enforcement of security and safety policy in commodity iot. In: NDSS (2019)
8. Chi, H., Fu, C., Zeng, Q., Du, X.: Delay wreaks havoc on your smart home: Delay-based automation interference attacks. In: 2022 IEEE Symposium on Security and Privacy (SP). pp. 285–302. IEEE (2022)
9. Community, P.: Tapo: Python interface for tp link tapo devices. `https://pypi.org/project/tapo/` (2025), online library accessed in 2025
10. Dong, Y., Li, Q., Wu, K., Li, R., Zhao, D., Tyson, G., Peng, J., Jiang, Y., Xia, S., Xu, M.: HorusEye: A realtime IoT malicious traffic detection framework using programmable switches. In: 32nd USENIX Security Symposium (USENIX Security 23). pp. 571–588. USENIX Association, Anaheim, CA (Aug 2023), `https://www.usenix.org/conference/usenixsecurity23/presentation/dong-yutao`
11. Fu, Z., Liu, M., Qin, Y., Zhang, J., Zou, Y., Yin, Q., Li, Q., Duan, H.: Encrypted malware traffic detection via graph-based network analysis. In: Proceedings of the 25th International Symposium on Research in Attacks, Intrusions and Defenses. pp. 495–509 (2022)
12. Girish, A., Hu, T., Prakash, V., Dubois, D.J., Matic, S., Huang, D.Y., Egelman, S., Reardon, J., Tapiador, J., Choffnes, D., et al.: In the room where it happens: Characterizing local communication and threats in smart homes. In: Proceedings of the 2023 ACM on Internet Measurement Conference. pp. 437–456 (2023)
13. Hu, T., Dubois, D.J., Choffnes, D.: Behaviot: Measuring smart home iot behavior using network-inferred behavior models. In: Proceedings of the 2023 ACM on Internet Measurement Conference. pp. 421–436 (2023)
14. Ikpehai, A., Adebisi, B., Anoh, K.: Effects of traffic characteristics on energy consumption of iot end devices in smart city. In: 2018 Global Information Infrastructure and Networking Symposium (GIIS). pp. 1–6 (2018). `https://doi.org/10.1109/GIIS.2018.8635744`
15. Jamshidi, S., Nafi, K.W., Nikanjam, A., Khomh, F.: Evaluating machine learning-driven intrusion detection systems in iot: Performance and energy consumption. Computers & Industrial Engineering **204**, 111103 (2025)
16. Jung, W., Zhao, H., Sun, M., Zhou, G.: Iot botnet detection via power consumption modeling. Smart Health **15**, 100103 (2020). `https://doi.org/https://doi.org/10.1016/j.smhl.2019.100103`, `https://www.sciencedirect.com/science/article/pii/S2352648319300674`
17. Kolias, C., Kambourakis, G., Stavrou, A., Voas, J.: Ddos in the iot: Mirai and other botnets. Computer **50**(7), 80–84 (2017)
18. Lightbody, D., Ngo, D.M., Temko, A., Murphy, C.C., Popovici, E.: Attacks on iot: side-channel power acquisition framework for intrusion detection. Future Internet **15**(5), 187 (2023)

19. Liu, H., Spink, T., Patras, P.: Uncovering security vulnerabilities in the belkin wemo home automation ecosystem. In: 2019 IEEE International Conference on Pervasive Computing and Communications Workshops (PerCom Workshops). pp. 894–899. IEEE (2019)
20. Mandalari, A.M., Haddadi, H., Dubois, D.J., Choffnes, D.: Protected or porous: A comparative analysis of threat detection capability of iot safeguards. In: 2023 IEEE Symposium on Security and Privacy (SP). pp. 3061–3078. IEEE (2023)
21. Merlino, V., Allegra, D.: Energy-based approach for attack detection in iot devices: A survey. Internet of Things **27**, 101306 (2024)
22. Mosenia, A., Jha, N.K.: A comprehensive study of security of internet-of-things. IEEE Transactions on emerging topics in computing **5**(4), 586–602 (2016)
23. Muhammad, K., Lloret, J., Singh, D., Kumari, S., Raj, R.: A survey on consumer internet of things (iot): A roadmap for iot adoption. IEEE Access **9**, 129793–129817 (2021)
24. Nimmy, K., Dilraj, M., Sankaran, S., Achuthan, K.: Leveraging power consumption for anomaly detection on iot devices in smart homes. Journal of Ambient Intelligence and Humanized Computing **14**(10), 14045–14056 (2023)
25. OConnor, T., Enck, W., Reaves, B.: Blinded and confused: uncovering systemic flaws in device telemetry for smart-home internet of things. In: Proceedings of the 12th Conference on Security and Privacy in Wireless and Mobile Networks. pp. 140–150 (2019)
26. Ren, J., Dubois, D.J., Choffnes, D., Mandalari, A.M., Kolcun, R., Haddadi, H.: Information exposure from consumer iot devices: A multidimensional, network-informed measurement approach. In: Proceedings of the Internet Measurement Conference. pp. 267–279 (2019)
27. Ronen, E., Shamir, A.: Extended functionality attacks on iot devices: The case of smart lights. In: 2016 IEEE European Symposium on Security and Privacy (EuroS&P). pp. 3–12. IEEE (2016)
28. Ronen, E., Shamir, A., Weingarten, A.O., O'Flynn, C.: Iot goes nuclear: Creating a zigbee chain reaction. In: 2017 IEEE Symposium on Security and Privacy (SP). pp. 195–212. IEEE (2017)
29. Ryan Smith, Daniel Palin, P.P.I.V.G.V., Shahandashti, S.F.: Battery draining attacks against edge computing nodes in iot networks. Cyber-Physical Systems **6**(2), 96–116 (2020). `https://doi.org/10.1080/23335777.2020.1716268`, `https://doi.org/10.1080/23335777.2020.1716268`
30. Safronov, V., Mandalari, A.M., Dubois, D.J., Choffnes, D., Haddadi, H.: Sunblock: Cloudless protection for iot systems. In: International Conference on Passive and Active Network Measurement. pp. 322–338. Springer (2024)
31. Sgueglia, A., Di Sorbo, A., Visaggio, C.A., Canfora, G.: A systematic literature review of iot time series anomaly detection solutions. Future Generation Computer Systems **134**, 170–186 (2022)
32. Syed, N.F., Baig, Z., Ibrahim, A., Valli, C.: Denial of service attack detection through machine learning for the iot. Journal of Information and Telecommunication **4**(4), 482–503 (2020)
33. Tahsien, S.M., Karimipour, H., Spachos, P.: Machine learning based solutions for security of internet of things (iot): A survey. Journal of Network and Computer Applications **161**, 102630 (2020). `https://doi.org/https://doi.org/10.1016/j.jnca.2020.102630`, `https://www.sciencedirect.com/science/article/pii/S1084804520301041`
34. Technologies, T.L.: Tapo p110 smart plug. `https://www.tapo.com/en/product/smart-plug/tapo-p110/`, accessed 2025
35. Trimananda, R., Varmarken, J., Markopoulou, A., Demsky, B.: Packet-level signatures for smart home devices. In: Network and Distributed Systems Security (NDSS) Symposium. vol. 2020 (2020)
36. Wan, Y., Xu, K., Xue, G., Wang, F.: Iotargos: A multi-layer security monitoring system for internet-of-things in smart homes. In: IEEE INFOCOM 2020-IEEE Conference on Computer Communications. pp. 874–883. IEEE (2020)
37. Zhang, W., Meng, Y., Liu, Y., Zhang, X., Zhang, Y., Zhu, H.: Homonit: Monitoring smart home apps from encrypted traffic. p. 1074–1088. CCS '18, Association for Computing Machinery, New York, NY, USA (2018). `https://doi.org/10.1145/3243734.3243820`, `https://doi.org/10.1145/3243734.3243820`

A Random Forest Power Analysis

In this section we report the raw accuracy and F1 numbers for each device individually from the setup described in Section 4.2.

Table 8: Per-device performance summary: overall, idle, and active metrics, per-attack macro F1, and MASC in the active state. Accuracy and F1 are reported in %, MASC is unitless.

Device		Overall		Idle state		Active state		Per-attack macro F1				$\text{MASC}_{\text{active}}$
Category	Name	Acc (%)	F1 (%)	Acc (%)	F1 (%)	Acc (%)	F1 (%)	ICMP (%)	OS (%)	SYN (%)	PORT (%)	
Appliances	Cosori Air Fryer	84.2	75.5	89.2	80.2	70.5	42.9	77.0	77.0	77.0	71.1	23254.2
	Honeywell Thermostat	94.2	92.2	96.6	95.6	92.6	89.3	93.0	94.5	88.5	93.0	19.8
	Levoit Air Purifier	94.7	92.8	96.6	95.2	93.2	88.5	97.0	98.6	85.3	90.2	52.0
	Swan Alexa Smart Kettle	91.2	87.1	94.9	92.5	95.5	93.6	87.6	82.0	94.2	84.7	34.4
	Weekett Kettle	87.6	82.1	90.9	86.3	84.1	75.6	85.8	78.3	73.5	90.7	15.6
	Xiaomi Blender	85.9	81.8	92.0	89.3	83.5	77.3	90.2	74.0	75.8	87.0	7056.4
Speakers	Bose	83.1	73.9	82.4	76.1	72.7	51.4	79.2	79.2	68.5	68.9	393.4
	Echo Dot 3	79.9	70.0	87.5	83.2	75.6	61.6	59.9	61.8	83.9	74.5	83.6
	Echo Dot 5	73.2	62.2	85.6	80.3	63.6	47.3	53.4	56.4	59.2	80.0	156.3
	Echo Spot	82.6	78.3	83.5	74.3	80.7	78.9	86.2	80.7	73.3	73.1	28.2
	Google Home	81.9	75.5	88.6	84.8	64.8	51.7	81.4	81.4	67.9	71.3	41.5
	Homepod	75.4	60.2	76.1	66.7	75.0	50.4	62.5	56.8	50.6	71.1	21.2
	Nest Mini 2	87.8	84.9	90.9	82.4	85.2	85.0	68.3	96.9	74.5	100.0	18.3
	Sonos	82.2	73.9	97.7	97.0	69.3	50.4	70.0	70.0	78.3	77.5	294.3
Cameras	Amcrest	72.2	60.3	80.7	71.5	72.7	56.7	62.0	52.9	45.4	81.1	25.9
	Blurams Security Cam	91.7	89.2	92.0	90.2	92.0	88.8	95.8	80.7	87.6	92.6	23.4
	Boifun Baby Cam	86.2	80.3	88.1	83.0	92.6	89.7	77.5	80.2	80.2	83.4	20.5
	Google Nest Cam	80.4	70.7	90.9	83.4	67.6	52.2	84.2	55.3	73.2	69.9	23.3
	Hualai Cam	98.3	97.8	96.6	95.5	97.2	96.3	98.6	97.2	95.6	100.0	21.3
	Yi Cam	84.2	76.5	94.9	92.9	88.1	78.0	72.2	80.1	67.4	86.2	13.8
Smart Bulbs	Antela Smart Bulb	80.8	75.2	88.6	84.4	72.7	61.1	94.3	68.9	59.7	77.9	32.2
	Fitop	97.0	96.0	96.6	95.5	98.3	97.7	98.5	97.1	92.6	95.6	25.2
	Lepro Smart Bulb	90.6	86.6	96.6	95.4	86.4	78.0	91.5	91.0	81.3	82.6	29.5
	Lifx Mini	92.8	90.5	96.0	95.0	88.6	84.0	94.0	89.0	84.5	94.5	36.4
	Tapo L530e	95.6	94.1	90.9	88.1	95.5	93.8	95.7	91.5	92.4	97.0	28.2
	Wiz Smart Bulb	89.7	85.5	90.9	87.7	90.9	86.6	91.3	85.4	74.5	91.0	28.1
	Yeelight	89.3	86.5	92.6	89.4	84.7	82.0	90.0	77.6	88.6	89.6	37.2
TV / Media	Apple TV A1625	86.6	82.7	92.4	90.7	79.9	72.9	85.6	71.7	79.9	93.8	29.2
	LG 32" TV	90.3	87.2	97.2	96.1	77.3	73.8	90.7	90.7	80.4	87.1	113.7
	Samsung Chromecast Smart TV	68.2	48.1	64.8	42.5	69.3	51.2	57.5	40.4	50.6	44.0	270.6
	Samsung 32" TV	75.3	58.0	90.9	84.7	64.2	41.3	64.8	46.6	66.4	54.3	151.7
	TCL Roku TV	74.5	56.7	87.5	78.2	68.8	47.6	49.9	56.7	71.1	49.2	214.8
	toshiba	80.0	72.3	92.0	88.4	69.3	53.1	72.3	80.0	69.5	67.5	74.2

B Per Attack Correlation Lists

We report the full Spearman correlation results for all active state background indicators and the per attack F1 scores in Table 9. Consolidating all four attack types into a single table provides a unified view of how background traffic behaviours align with detection performance across SYN flooding, ICMP flooding, Port Scan and OS Fingerprinting. The table presents the dominant indicators for each attack type, allowing direct comparison of which background properties suppress or reinforce detectability for different adversarial behaviours. Each entry lists the Spearman coefficient r together with its p value, which together express both the direction and the statistical strength of the association.

Spearman correlation is used because many background indicators vary in a monotonic but non linear manner across devices, particularly under active workloads where traffic patterns can span several orders of magnitude. Rank based correlation therefore provides a stable measure of association that is robust to heavy-tailed packet volumes, protocol skew and large inter device variability. The combined table also highlights the extent to which masking arises from different mechanisms. Some indicators, such as mDNS, burst frequency and packet size variance, consistently correlate with reduced detectability, while others, such as DHCP or IAT related metrics, occasionally correlate with improved separation.

Table 9: Correlation results for all attack types

SYN Flood			**ICMP Flood**			**Port Scanning**			**OS Fingerprinting**		
Feature	**r**	**p**	**Feature**	**r**	**p**	**Feature**	**r**	**p**	**Feature**	**r**	**p**
mDNS	-0.47	0.005256	mDNS	-0.70	0.000005	size mean	-0.57	0.000529	BTS	-0.45	0.008845
BTS	-0.45	0.008889	dst IP spread	-0.67	0.000019	TLS	-0.56	0.000696	DHCP	0.45	0.009428
inout ratio	-0.45	0.009004	TLS	-0.60	0.000248	burst	-0.56	0.000697	mDNS	-0.44	0.009523
DHCP	0.45	0.009149	BTS	-0.59	0.000291	dst IP spread	-0.55	0.001028	TLS	-0.43	0.011830
TLS	-0.44	0.010544	inout ratio	-0.59	0.000300	other protocol	0.54	0.001122	PPS	-0.43	0.012021
PPS	-0.44	0.010577	PPS variance	-0.59	0.000306	SSDP	-0.48	0.004392	IAT mean	0.43	0.012021
IAT mean	0.44	0.010577	dst port spread	-0.57	0.000478	inout ratio	-0.45	0.008531	packet count	-0.43	0.012021
packet count	-0.44	0.010577	burst	-0.57	0.000502	packet count	-0.44	0.010784	inout ratio	-0.42	0.014234
dst IP spread	-0.42	0.015864	size mean	-0.57	0.000610	IAT mean	0.44	0.010784	size mean	-0.41	0.016875
burst	-0.42	0.016058	size var	-0.56	0.000656	PPS	-0.44	0.010784	dst IP spread	-0.41	0.018942